Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 5 · SLO 1 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 5 : SLO â€“ 1 | Infix to Postfix Conversion â€” Worked Example

I. Step-by-Step Trace: A - (B / C + (D % E * F) / G) * H

• A -> operand | Output: A | Stack: -
• - -> push | Output: A | Stack: -
• (-> push | Output: A | Stack: - (
• B -> operand | Output: AB | Stack: - (
• / -> push | Output: AB | Stack: - (/
• C -> operand | Output: ABC | Stack: - (/
• + -> pop /, push + | Output: ABC/ | Stack: - (+
• (-> push | Output: ABC/ | Stack: - (+ (
• D -> operand | Output: ABC/D | Stack: - (+ (
• % -> push | Output: ABC/D | Stack: - (+ (%
• E -> operand | Output: ABC/DE | Stack: - (+ (%
• * -> pop %, push * | Output: ABC/DE% | Stack: - (+ (*
• F -> operand | Output: ABC/DE%F | Stack: - (+ (*
•) -> pop till (| Output: ABC/DE%F* | Stack: - (+
• / -> pop +, push / | Output: ABC/DE%F*+ | Stack: - (/
• G -> operand | Output: ABC/DE%F*+G | Stack: - (/
•) -> pop till (| Output: ABC/DE%F*+G/ | Stack: -
• * -> push | Output: ABC/DE%F*+G/ | Stack: - *
• H -> operand | Output: ABC/DE%F*+G/H | Stack: - *
• END -> pop * , pop - | Output: ABC/DE%F*+G/H*- | Stack: empty

II. Final Answer

• Postfix expression: A B C / D E % F * G / + H * -
• Compact form: ABC/DE%F*+G/H*-